

HarmonyXR Gaze-Pose Context System Project Completion and Status Report

HarmonyXR Gaze-Pose Context System

Project Completion and Status Report

Report Date: May 15, 2026

Project: HarmonyXR Gaze-Pose Context System

Prototype: Adaptive XR Context Prototype

Current Branch: `research/gaze-pose-context-communication-v0`

Primary Scope: Unity XR proof-of-concept for context inference, adaptive guidance, headset evidence, and research communication.

1. Executive Summary

The HarmonyXR Gaze-Pose Context proof-of-concept has progressed from a technical Unity XR prototype into a more polished, research-driven adaptive XR demonstration. The current system runs as a headset-based object-sorting experience and communicates the research purpose more clearly through onboarding, live guidance, adaptive support messages, runtime QA evidence, and a completion summary.

The prototype demonstrates that multimodal XR signals can be captured during a task and used to infer user context states such as Engaged, Distracted, Transitioning, and Idle. Based on these inferred states, the interface adapts user-facing guidance and support without changing the approved research scope or adding unrelated gameplay systems.

The current project is ready for internal review, stakeholder demonstration, and paper-supporting evidence use. Remaining work is mainly final presentation validation, paper refinement, and any supervisor-requested documentation adjustments.

2. Project Objective

The main objective of this work was to improve the Unity XR proof-of-concept so first-time users, clients, supervisors, and stakeholders can clearly understand:

- what the application is,
- why the user is performing the task,
- what the system is detecting,
- what research problem is being addressed,
- and why adaptive XR behavior matters.

The work stayed within the approved proof-of-concept boundary. No new game-play loop, unrelated study system, mobile-phone mode, or expanded research architecture was added.

3. Initial Project State

At the start of this improvement cycle, the core technical pipeline already existed:

- TrainingSimulation object-sorting task,
- context inference pipeline,
- four context states: Engaged, Distracted, Transitioning, and Idle,
- user guide panel,
- adaptive behavior support,
- debug/QA overlay,
- context logging.

However, the experience still felt closer to a technical Unity prototype than a polished research demonstration. The main gaps were communication, readability, stakeholder confidence, and presentation quality.

Key issues identified:

- onboarding explained the task more than the research purpose,
 - users could understand sorting but not the value of multimodal context inference,
 - UI panels were visually dense or debug-like,
 - completion feedback did not strongly communicate research success,
 - QA evidence was useful but needed clearer export and interpretation,
 - headset demo visuals were still plain due to the default skybox.
-

4. Work Completed

4.1 Onboarding and Research Communication

The onboarding flow was improved to explain the application as an adaptive XR research prototype rather than only a sorting task.

Completed improvements:

- added clearer research-purpose language,
- improved explanation of the four context states,
- refined instructional wording for professional tone,
- made onboarding progression user-controlled,
- reduced dense paragraph-style guidance,
- improved visual hierarchy and spacing,
- aligned the guide panel more comfortably for headset viewing.

Current result:

The user is introduced to the experience step by step and can understand that the sorting task is used to create observable behavior for context inference.

4.2 Adaptive XR Guidance

The prototype now communicates adaptive behavior more clearly during runtime.

Implemented adaptive behavior communication:

- Engaged: continues normal task guidance when the user appears focused,
- Distracted: presents refocus or attention guidance,
- Transitioning: gives soft guidance while the user moves between task steps or targets,
- Idle: presents adaptive support actions such as Continue, Take a break, and Recenter view,
- Incorrect selection: shows corrective guidance when the wrong object is selected,
- Completion: presents a research-focused summary after all objects are sorted.

Current result:

The adaptive behavior is visible to users and stakeholders as runtime guidance and support, not only as hidden internal state logic.

4.3 Object Task Feedback

The object-sorting task was refined so the user receives clearer feedback when the active object does not match the selected object.

Completed improvements:

- added incorrect-object selection messaging,
- reinforced the active task object,
- helped users recover without changing the task design,
- preserved the simple research-controlled interaction flow.

Current result:

The task is easier to follow, and incorrect selections now support the research narrative by showing context-aware guidance.

4.4 QA and Runtime Evidence Logging

Runtime QA evidence was improved so important context values can be reviewed from headset console logs.

Captured QA fields include:

- STATE,
- CONF,
- BOUND,
- AOI,
- POSTURE,
- FIX,
- DWELL,
- SACC,
- BODY,
- HAND,
- PINCH,
- DIST.

Current result:

The system produces runtime QA evidence that supports research validation and explanation of the context inference pipeline.

4.5 QA Panel and Developer Presentation

The QA/Event panel behavior was refined to reduce the debug-build feeling during normal user interaction.

Completed improvements:

- avoided showing the QA panel for simple pinch actions alone,
- repositioned the panel away from the main interaction area,
- improved terminology from developer/debug language toward research-oriented language,
- kept QA information available for evidence and developer explanation.

Current result:

The normal user experience is cleaner, while QA mode remains useful for demonstrating internal evidence when required.

4.6 End-of-Session Completion Experience

The final completion screen was redesigned to make the result clearer and more professional.

Completed improvements:

- improved spacing and hierarchy,
- added stronger task-completion messaging,
- removed repeated or merged text,
- highlighted adaptive XR context analysis completion,
- presented a clearer research proof statement.

Current result:

The end of the experience now communicates that the task was completed and that adaptive XR context analysis was demonstrated.

4.7 Headset Evidence Collection

A combined headset evidence run was completed and organized.

Evidence collected:

- full headset evidence video,
- Unity/ADB console evidence,
- filtered QA evidence,
- processed QA metrics table,
- state coverage table,
- adaptation event table,
- stability/latency summary table,
- evidence summary documentation.

Verified context-state coverage:

State	Observed	Count	First Timestamp	Last Timestamp
Engaged	Yes	15	05-14 16:19:18.956	05-14 16:21:33.401
Distracted	Yes	30	05-14 16:16:46.104	05-14 16:21:55.453
Transitioning	Yes	100	05-14 16:19:28.044	05-14 16:21:48.434
Idle	Yes	24	05-14 16:16:49.126	05-14 16:22:02.470

Current result:

The evidence package supports the claim that all four context states were observed during headset execution.

4.8 Research Paper Draft

A Lavanya-focused research paper draft was prepared using current implementation details and collected evidence.

Completed outputs:

- Markdown paper source,
- DOCX paper draft,
- PDF paper draft,
- supporting figures,
- Phase 4/user study treated only as future work.

Current result:

The paper draft is available for review and further refinement. It does not claim completed user-study results.

4.9 Demo Presentation Preparation

The project was prepared for presentation through script and visual-polish work.

Completed improvements:

- prepared a LinkedIn/company demo voiceover script,
- aligned the script with the actual headset flow,
- described onboarding, task execution, incorrect selection, adaptive support, evidence logging, and completion,
- added a custom neutral demo skybox to replace the Unity default skybox in the TrainingSimulation scene.

Current result:

The demo is better prepared for video recording and stakeholder presentation. The new skybox should be visually checked in Unity/headset before final sharing.

5. Evidence Files

Primary evidence files currently available:

- Full headset evidence video: 20260514_run01_full-evidence.mp4
- Full console evidence: 20260514_run01_console-evidence.txt
- Filtered QA evidence: 20260514_run01_filtered-evidence.txt
- QA metrics table: qa_metrics_extract.csv
- State coverage table: state_coverage_table.csv
- Adaptation event table: adaptation_event_table.csv
- Stability/latency table: stability_latency_table.csv
- Evidence summary: EVIDENCE_SUMMARY_20260514.md

These files are stored under the **Research_Evidence** folder using the organized evidence structure created for the project.

6. Current Project Status

Current status:

The approved Unity XR proof-of-concept implementation is functionally complete for the current research-demonstration scope and is ready for review, demo recording, and paper-supporting use.

Reason for this status:

- the headset experience runs and supports the sorting task,
- onboarding and guidance now communicate the research purpose,
- the system demonstrates four context states,
- adaptive guidance is visible during runtime,
- incorrect selection and idle support behavior are represented,
- completion screen communicates research success,
- QA metrics are captured in headset logs,
- all four context states are present in processed evidence,
- paper draft and evidence package are prepared,
- visual presentation has been improved with a custom demo skybox.

This status does not mean the full research program is complete. It means the current proof-of-concept and evidence package are complete enough for supervisor review, stakeholder demonstration, and paper-draft support.

7. Remaining Work

Remaining work is limited to review, validation, and final presentation tasks:

- validate the new skybox visually in Unity and headset,
 - record the final demo video using the prepared script,
 - review the paper draft with supervisor/team feedback,
 - update the paper if requested,
 - avoid claiming Phase 4 user-study outcomes until that work is actually conducted,
 - keep evidence files organized for final submission and presentation.
-

8. Important Limitations

The following limitations should be stated clearly when presenting the project:

- This is a proof-of-concept, not a completed large-scale user study.
 - Phase 4/user-study work is future work unless separately completed.
 - The adaptive behavior is intentionally lightweight and focused on guidance/support.
 - The system adapts UI guidance and feedback; it does not introduce unrelated gameplay or autonomous task redesign.
 - `room_scale` posture values should be interpreted as spatial/boundary context, not strict sitting/standing classification.
 - The current evidence supports runtime state coverage and signal logging, but formal statistical user validation remains outside the completed scope.
-

9. Final Assessment

The project has successfully moved from a technically working Unity XR prototype to a clearer, more presentable adaptive XR research proof-of-concept.

The current implementation demonstrates:

- headset-based task execution,
- multimodal XR signal capture,
- context-state inference,
- adaptive runtime guidance,
- corrective task feedback,
- idle support behavior,
- research-focused completion summary,
- QA evidence logging,
- organized evidence package,
- and presentation-ready documentation outputs.

The project is currently suitable for final demo preparation, internal team review, stakeholder explanation, and research-paper refinement within the approved scope.